



Cerberus - How Radix achieves infinite linear scalability while preserving atomic composability

Introduction

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At the core of Radix lies its unique consensus protocol, Cerberus. Cerberus will have two distinct features that no other decentralized Distributed Ledger Technology (DLT) has yet to achieve: (1) practically infinite linear scalability; while preserving (2) cross-shard atomic composability.

Cerberus achieves this by starting with a unique data structure that pre-shards the ledger into 2^{256} shards [1]. This is large enough to fit every thousandth atom in the observable universe into its own shard; or for another comparison, all possible combinations of Bitcoin addresses could fit in the Cerberus “shardspace” 79 billion billion billion times. Every change to the ledger, or “substate”, is deterministically allocated its own shard based upon its hash.

When a transaction occurs, Cerberus’ novel consensus design allows nodes to temporarily “braid” consensus across the shards of relevant substates together. Related substates can thus be composed into atomic transactions when needed, and unrelated substates can be processed completely in parallel. Each node is only required to serve a subset of shards - no global state or ledger is maintained by any one node.

Because of this, as nodes are added to the network, transaction throughput increases linearly without practical limit; transactions reach settlement finality in less than five seconds; transaction fees will always be tiny; nodes can always be run on simple hardware; and the ability to compose transactions atomically across the global ledger will never be sacrificed.

By the end of this infographic series you will understand what this means and how it works. A more in depth and technical description of Cerberus can be found in the Cerberus whitepaper [here](#); and an independent validation of the safety and liveness of Cerberus by the University of California Davis can be found [here](#).

Unique Features



Practicaly Infinite Linear Scalability

Allows Radix to always scale to meet the needs of network usage – just like the internet, there is no ceiling



Cross-Shard Atomic Composability

Allows transactions to be composed across shards - either settling all together, or not at all - across the global ledger

[1] Shards in Cerberus: 115,792,089,237,316,195,423,570,985,008,687,907,853,269,984,665,640,564,039,457,584,007,913,129,639,936

The version of Cerberus described in this infographic series is scheduled to launch as part of the fully sharded Radix Xi'an release. Please visit www.radixdlt.com for details on the Radix roadmap.